Use Case Descriptions

Hardware Market E-Commerce System

وصف حالات الاستخدام — نظام متجر إلكتروني للهاردوير

Detailed Use Case Descriptions

This document provides detailed descriptions for the main Use Cases of the Hardware Market E-Commerce System. Each Use Case includes actors, pre-conditions, post-conditions, main flow, alternative flows, and exception flows.

UC-01: Browse Products

Use Case ID	UC-01
Use Case Name	Browse Products
Actor(s)	Guest, Customer
Description	Allows the user to view the list of available products in the store.
Pre-conditions	The system is running normally. Products exist in the database.
Post-conditions	A list of products is displayed to the user.

Main Flow (Basic Flow):

1. The user navigates to the home page or products page.
2. The system retrieves the list of products from the database.
3. The system displays products with thumbnail image, name, price, and stock status.
4. The user can scroll or navigate between pages if pagination exists.

Alternative Flows:

• No products available: The system displays a message "No products available at the moment".

Exception Flows:

• Database connection error: The system displays a general error message.

UC-02: Search & Filter Products

Use Case ID	UC-02
Use Case Name	Search & Filter Products
Actor(s)	Guest, Customer
Description	Allows the user to search for products using keywords or filter them by category, price, or stock status.
Pre-conditions	The system is running. Products exist in the database.
Post-conditions	Matching products are displayed according to the search/filter criteria.

Main Flow (Basic Flow):

1. The user enters a search keyword or selects filter criteria (category, price range, in-stock only).
2. The user clicks the Search or Apply Filter button.
3. The system queries the database based on the criteria.
4. The system displays the matching products.

Alternative Flows:

• No matching results: The system displays "No products found matching your criteria".

Exception Flows:

• Database error: The system displays an error message.

UC-03: View Product Details

Use Case ID	UC-03
Use Case Name	View Product Details
Actor(s)	Guest, Customer
Description	Displays the full details of a selected product.
Pre-conditions	The product exists in the system.
Post-conditions	The product details page is displayed.

Main Flow (Basic Flow):

1. The user clicks on a product from the product list.
2. The system retrieves the full product information.
3. The system displays: image, name, price, description, specifications, available quantity, and stock status (In Stock / Out of Stock).

Alternative Flows:

• Product is out of stock: The status is clearly shown as "Out of Stock" and the Add to Cart button is disabled.

Exception Flows:

• Product not found (deleted): The system displays "Product not available".

UC-04: Register Account

Use Case ID	UC-04
Use Case Name	Register Account
Actor(s)	Guest
Description	Allows a visitor to create a new customer account.
Pre-conditions	The user is not logged in.
Post-conditions	A new account is created in the system. The user may be logged in automatically or redirected to the login page.

Main Flow (Basic Flow):

1. The user clicks on "Register" or "Create Account".
2. The system displays the registration form (name, email, password, confirm password, phone, address).
3. The user fills in the required fields and clicks "Register".
4. The system validates the entered data.
5. The system creates the new account and displays a success message.

Alternative Flows:

• Email already registered: The system displays "This email is already registered".
• Password and confirmation do not match: The system displays an appropriate error message.

Exception Flows:

• Missing or invalid data: The system highlights the invalid fields with error messages.

UC-05: Switch Language

Use Case ID	UC-05
Use Case Name	Switch Language
Actor(s)	Guest, Customer, Admin
Description	Allows the user to switch the interface language between Arabic and English at any time.
Pre-conditions	None.
Post-conditions	The interface language is changed according to the user's selection.

Main Flow (Basic Flow):

1. The user clicks the language switch button (العربية / English).
2. The system changes all interface texts to the selected language immediately.

Alternative Flows:

• None.

Exception Flows:

• None.

UC-06: Login

Use Case ID	UC-06
Use Case Name	Login
Actor(s)	Customer, Admin
Description	Allows a registered user to log into the system.
Pre-conditions	The user has a registered account.
Post-conditions	The user is authenticated and redirected to the appropriate page (Customer home or Admin dashboard).

Main Flow (Basic Flow):

1. The user navigates to the Login page.
2. The user enters email and password.
3. The user clicks the Login button.
4. The system verifies the credentials.
5. If valid, the system creates a session and redirects the user.

Alternative Flows:

• Invalid credentials: The system displays "Invalid email or password".

Exception Flows:

• Account does not exist: The system displays an error message.

UC-08: Add to Cart

Use Case ID	UC-08
Use Case Name	Add to Cart
Actor(s)	Customer
Description	Allows the customer to add a product to the shopping cart if it is available in stock.
Pre-conditions	The customer is logged in. The product is in stock.
Post-conditions	The product is added to the customer's cart.

Main Flow (Basic Flow):

1. The customer views a product details page.
2. The customer selects the desired quantity.
3. The customer clicks "Add to Cart".
4. The system checks stock availability.
5. If available, the system adds the product to the cart and displays a confirmation message.

Alternative Flows:

• Product is out of stock: The Add to Cart button is disabled or the system shows "Out of Stock".
• Requested quantity exceeds available stock: The system displays a warning message.

Exception Flows:

• User not logged in: The system redirects to the Login page.

UC-09: Manage Cart

Use Case ID	UC-09
Use Case Name	Manage Cart
Actor(s)	Customer
Description	Allows the customer to view, update quantities, or remove items from the shopping cart.
Pre-conditions	The customer is logged in. The cart contains at least one item (or is empty).
Post-conditions	The cart is updated according to the customer's actions.

Main Flow (Basic Flow):

1. The customer opens the shopping cart page.
2. The system displays all items in the cart with name, price, quantity, and subtotal.
3. The customer can increase/decrease quantity or remove an item.
4. The system updates the cart and recalculates the total.

Alternative Flows:

• Cart is empty: The system displays "Your cart is empty".

Exception Flows:

• Product stock changed while in cart: The system notifies the user and adjusts the quantity.

UC-10: Place Order (Checkout)

Use Case ID	UC-10
Use Case Name	Place Order (Checkout)
Actor(s)	Customer
Description	Allows the customer to complete the purchase by providing shipping information and confirming the order.
Pre-conditions	The customer is logged in. The cart contains at least one in-stock product.
Post-conditions	A new order is created with status "Processing". Cart is cleared. Stock quantities are updated.

Main Flow (Basic Flow):

1. The customer clicks "Proceed to Checkout" from the cart.
2. The system displays the checkout form (shipping address, phone, notes).
3. The customer fills in the shipping details and confirms the order.
4. The system validates the data and checks stock one final time.
5. The system creates the order, updates product stock, clears the cart, and displays a success message with order number.

Alternative Flows:

• A product became out of stock during checkout: The system removes it or notifies the customer and prevents completing the order.

Exception Flows:

• Missing shipping information: The system highlights required fields.
• Database error during order creation: The system displays an error and does not deduct stock.

UC-11: View Order History

Use Case ID	UC-11
Use Case Name	View Order History
Actor(s)	Customer
Description	Allows the customer to view their previous orders and the status of each order.
Pre-conditions	The customer is logged in.
Post-conditions	A list of the customer's orders is displayed.

Main Flow (Basic Flow):

1. The customer navigates to "My Orders" page.
2. The system retrieves all orders belonging to the customer.
3. The system displays order number, date, total amount, and current status.

Alternative Flows:

• No previous orders: The system displays "You have no orders yet".

Exception Flows:

• Database error: The system displays a general error message.

UC-14: Manage Products

Use Case ID	UC-14
Use Case Name	Manage Products
Actor(s)	Admin
Description	Allows the admin to add, edit, or delete products in the system.
Pre-conditions	The admin is logged in.
Post-conditions	Product data is created, updated, or deleted successfully.

Main Flow (Basic Flow):

1. The admin navigates to the Products management page.
2. To Add: Admin fills product form (name, description, price, quantity, category, image) and saves.
3. To Edit: Admin selects a product, modifies the data, and saves.
4. To Delete: Admin selects a product and confirms deletion.
5. The system updates the database and displays a confirmation message.

Alternative Flows:

• Invalid data entered: The system shows validation errors.

Exception Flows:

• Image upload failure: The system displays an error and does not save the product.

UC-16: Manage Orders

Use Case ID	UC-16
Use Case Name	Manage Orders
Actor(s)	Admin
Description	Allows the admin to view all orders and change their status.
Pre-conditions	The admin is logged in.
Post-conditions	Order status is updated successfully.

Main Flow (Basic Flow):

1. The admin navigates to the Orders management page.
2. The system displays a list of all orders with details.
3. The admin selects an order and changes its status (Processing / Shipped / Delivered / Canceled).
4. The system updates the order status and saves the change.

Alternative Flows:

• No orders exist: The system displays "No orders found".

Exception Flows:

• Database update failure: The system displays an error message.

UC-17: Update Stock

Use Case ID	UC-17
Use Case Name	Update Stock
Actor(s)	Admin
Description	Allows the admin to update the available quantity of products.
Pre-conditions	The admin is logged in. The product exists.
Post-conditions	Product stock quantity is updated. Status changes to Out of Stock if quantity reaches zero.

Main Flow (Basic Flow):

1. The admin opens a product for editing or uses a quick stock update feature.
2. The admin enters the new quantity.
3. The system saves the new quantity.
4. If quantity = 0, the system automatically sets status to Out of Stock.

Alternative Flows:

• Quantity set to zero: Status automatically becomes Out of Stock.

Exception Flows:

• Invalid quantity (negative number): The system rejects the input.